



Concomitant administration of *Moringa oleifera* seed powder in the remediation of arsenic-induced oxidative stress in mouse

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Abstract

Contamination of ground water by arsenic has become a cause of global public health concern. In West Bengal, India, almost 6 million people are endemically exposed to inorganic arsenic by drinking heavily contaminated groundwater through hand-pumped tube wells. No safe, effective and specific preventive or therapeutic measures for treating arsenic poisoning are available. We recently reported that some of the herbal extracts possess properties effective in reducing arsenic concentration and in restoring some of the toxic effects of arsenic in animal models. *Moringa oleifera* Lamarck (English: Horseradish-tree, Drumstick-tree, Hindi: Saijan, Sanskrit: Shigru) belongs to the Moringaceae family, is generally known in the developing world as a vegetable, a medicinal plant and a source of vegetable oil.

The objective of the present study was to determine whether *Moringa oleifera* (*M. oleifera*) seed powder could restore arsenic induced oxidative stress and reduce body arsenic burden. Exposure to arsenic (2.5 mg/kg, intraperitoneally for 6 weeks) led to a significant increase in the levels of tissue reactive oxygen species (ROS), metallothionein (MT) and thiobarbituric acid reactive substance (TBARS) which were accompanied by a decrease in the activities in the antioxidant enzymes such as superoxide dismutase (SOD), catalase and glutathione peroxidase (GPx) in mice. Arsenic exposed mice also exhibited liver injury as reflected by reduced acid phosphatase (ACP), alkaline phosphatase (ALP) and aspartate aminotransferase (AST) activities and altered heme synthesis pathway as shown by inhibited blood δ -aminolevulinic acid dehydratase (δ -ALAD) activity.

Co-administration of *M. oleifera* seed powder (250 and 500 mg/kg, orally) with arsenic significantly increased the activities of SOD, catalase, GPx with elevation in reduced GSH level in tissues (liver, kidney and brain). These changes were accompanied by approximately 57%, 64% and 17% decrease in blood ROS, liver metallothionein (MT) and lipid peroxidation respectively in animal co-administered with *M. oleifera* and arsenic. Another interesting observation has been the reduced uptake of arsenic in soft tissues (55% in blood, 65% in liver, 54% in kidneys and 34% in brain) following administration of *M. oleifera* seed powder (particularly at the dose of 500 mg/kg). It can thus be concluded from the present study that concomitant administration of *M. oleifera* seed powder with arsenic could significantly protect animals from oxidative stress and in reducing tissue arsenic concentration. Administration of *M. oleifera* seed powder thus could also be beneficial during chelation therapy with a thiol chelator.

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1. Introduction

Arsenic is a ubiquitous element in the environment and has been classified as a group 1 carcinogen. The adverse health

effects caused by excessive arsenic exposure particularly from natural sources have been a major environmental health issue of global concern (Ng et al., 2003). Arsenic induced skin lesions are characterized by symptoms like hyper-pigmentation and keratosis. Other clinical manifestations include Blackfoot disease (Tseng, 1989), diabetes mellitus (Lia et al., 1994), hypertension (Chen et al., 1995), atherosclerosis

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(Simeonova and Luster, 2004), and cancers of the skin, lung, bladder and liver (Chen et al., 1992). The most serious consequence of chronic arsenic exposure is its carcinogenic effect. It has been at the top of the list of 20 hazardous substances released by the Agency for Toxic Substances and Disease Registry (ATSDR) and the United States Environmental Protection Agency (U.S.E.P.A.) (ATSDR, 1997).

The current U.S.E.P.A standard for arsenic contents in water is 10 µg/L. Chronic endemic arsenicosis was first reported in Bangladesh and India in 1996 (Guha Mazumder et al., 1999). Treatment for arsenic poisoning is restricted to some thiol containing chelating agents like meso 2,3-dimercaptosuccinic acid (DMSA), 2,3 dimercaptopropane-1-sulfonate (DMPS) or British Anti Lewisite (BAL; 2,3-dimercaprol) (Flora et al., 1995; Aposhian and Aposhian, 1990). They can be administered either alone or in combination with antioxidants like Vitamin C, Vitamin E (Kannan and Flora, 2004), zinc (Modi et al., 2005), N-acetyl cysteine (Flora, 1999), thiamine or methionine (Nandi et al., 2005). The clinical applications of these chelators are compromised by serious drawbacks like low therapeutic index, non-specificity (essential metal loss such as Cu and Zn), inability to penetrate cellular membrane (extracellular nature) and metal redistribution (Flora and Mehta, 2003). The other side effects include hepatotoxicity, renal toxicity, headache, nausea, vomiting, blood pressure lachrymation, profuse sweating, intense pain in the chest and abdomen and anxiety, gastrointestinal discomfort, skin reaction, mild neutropenia and embryo and fetal toxicity (Bosque et al., 1994).

Thus, there is a need for a safe and effective treatment for arsenic that should fulfill two criteria: (i) to prevent accumulation of arsenic in the target tissues; and (ii) to restore arsenic induced alterations in biochemical/clinical variables including oxidative stress. One of the strategies to achieve these goals could be by exploring the efficacies of herbal extracts (Kalia and Flora, 2005). 'Ayurveda' is an ancient form of Indian medicine, which deals with plants and plants products. This indigenous form of medicines uses the active ingredients present in plants to treat diseases. Plants are generally considered to be less toxic and free from many of the side effects that a synthetic drug may exhibit (Valiathan, 1998).

Moringa oleifera (English: Horseradish tree, Drumstick tree, Sanskrit: Shigru) belongs to a monogenic family of shrubs and trees, moringaceae. The trees originated from Agra and Oudh in North Western region of India to South of the Himalayan Mountains (Ramachandran et al., 1980). A native tree of northern India, is now widely cultivated throughout the tropics and is found in many countries of Africa, Asia and South America regions. Leaves of this plant are traditionally reported to possess various biological activities, including hypocholesterolemic, antidiabetic, hypotensive agent and anti-tumor agents (Mehta et al., 2003; Kar et al., 2003; Faizi et al., 1995; Guevara et al., 1999), regulate thyroid hormone (Tahiliani and Kar, 2000) and treat gastric ulcers (Pal et al., 1995). The leaves, as well as the flowers, roots, gums, fruits and seeds are extensively used for treating inflammation (Ezeamuzle et al., 1996), cardiovascular action (Limaye

et al., 1995), liver disease (Rao and Misra, 1998), cancer (Costa-Lotufo et al., 2005) and hematological, hepatic and renal function (Mazumder et al., 1999).

Beside this, *Moringa oleifera* (*M. oleifera*) seeds contain flocculants to purify water (McConnachie et al., 1999), antimicrobial substances (Filgona, 2005) and edible oils. Seed kernels of *M. oleifera* are known to remove lead, iron and cadmium ions from contaminated water (Sajidu et al., 2005). Using sorption studies, Sharma et al. (2006) reported that seed powder of *M. oleifera* has the ability to remove cadmium from aqueous system. However, its efficacy in protecting arsenic toxicity *in vivo* has not been studied in detail. We recently reported that animals treated with *M. oleifera*, post arsenic exposure, showed moderate recovery in the altered biochemical variables and in reducing blood and liver arsenic concentrations (Gupta et al., 2005).

In the present study we studied the efficacy of concomitant administration *M. oleifera* with arsenic in protecting mice from the toxic effects of arsenic (including oxidative stress) and preventing arsenic absorption in target tissues. Since exposure to arsenic might lead to redistribution of essential metals, the concentration of essential metals like zinc, copper and magnesium was determined. The imbalance in the concentration of some of these essential metals (particularly zinc) might affect the level of metallothionein, a cysteine rich cytosolic protein of low molecular weight.

2. Materials and methods

2.1. Chemicals

Sodium arsenite (Sigma Chemical, USA), δ-aminolevulinic acid (Sigma Chemical, USA) and all other chemicals were of analytical grade and purchased from Merck (Germany), BDH Chemical (Mumbai, India) or Sigma (USA).

2.2. Plant material

The *M. oleifera* seed powder was supplied by the Chemistry Division of Dayalbagh Educational Institute, Agra, India. The seeds were freshly collected and identified at the Chemistry Division of Dayalbagh Educational Institute, Agra, India. They were cut into small pieces and air dried indoor under subdued light and with good ventilation. They were then crushed into a coarse powder using a laboratory homogenizer. This coarse powder of *M. oleifera* was used as a plant material.

2.3. Animals and treatments

Male Swiss mice weighing approximately 20–25 g were obtained from the animal facility of Defence Research and Development Establishment (DRDE). The animals were acclimatized for 7 days prior to use. All animals received humane care in compliance with the guidelines of the "Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA)". Animal ethical committee of DRDE, Gwalior, India also approved the protocols for the experiments. Throughout the experiment, animals were housed in stainless steel cages in an air-conditioned room with temperature maintained at 25 ± 2 °C and 12 h with alternate day and night cycles. Mice were allowed standard pellet diet (Amrut Feeds, Pranav Agro, New Delhi, metal contents of diet, in ppm dry weight Zn 45, Cu 10, Mn 55, Fe 70, Co 5), and water *ad libitum*.

2.4. Experimental protocols

25 animals were randomized into five groups of five mice each and were treated as below for six weeks:

Group I: No treatment (normal animals).

Group II: 500 mg/kg *M. oleifera* seed powder, administered orally, once, daily.

Group III: 2.5 mg/kg arsenic (as sodium arsenite) administered intraperitoneally (i.p.), once, daily.

Group IV: Arsenic (as in group III) +250 mg/kg *M. oleifera*, orally, once daily.

Group V: Arsenic (as in group III) +500 mg/kg, *M. oleifera*, orally, once daily.

The seed powder of *M. oleifera* was given 1 h prior to the arsenic dose. The doses were selected on the basis of previously published reports suggesting that *M. oleifera* is non-toxic to humans and to animals (Gupta et al., 2005; Grabow et al., 1985). The food and water intake was recorded and mice were weighed every week. After the administration of the last dose, the animals were given 2 days' rest and were sacrificed under light ether anesthesia. Blood was collected by cardiac puncture in heparinized tubes. Liver, brain and kidneys were removed, rinsed in cold saline, blotted, weighed and used for various biochemical variables and metal analysis. Half portion of the liver, brain and one of the kidneys from each mouse were processed immediately for biochemical estimation and the remaining was stored at -20°C before wet acid digestion with HNO_3 for estimation of arsenic contents.

2.5. Biochemical assay

2.5.1. Blood δ -aminolevulinic acid dehydratase (ALAD) (EC 4.2.1.24)

The activity of blood δ -aminolevulinic acid dehydratase (ALAD) was assayed according to the procedure of Berlin and Schaller (1974). Total volume of 0.2 ml of heparinized blood was mixed with 1.3 ml of distilled water and incubated for 10 min at 37°C for complete hemolysis. After adding 1 ml of standard ALA, the tubes were incubated for 60 min at 37°C . The reaction was stopped after 1 h by adding 1 ml of trichloroacetic acid (TCA). After centrifugation equal volume of Ehrlich reagent (0.59 g of dimethylaminobenzaldehyde in 12.5 ml glacial acetic acid +6 ml perchloric acid +1 ml 2.5% mercuric chloride) was added and the absorbance was recorded at 555 nm after 5 min.

2.5.2. Urinary aminolevulinic acid (U-ALA)

Urinary δ -aminolevulinic acid was estimated as described by the method of Davis et al., (1968). Twenty-four hour samples were collected in disposable polypropylene tubes (acidified with a drop of acetic acid) and stored at 0°C until analyzed. The separation and isolation δ -ALA was performed by employing a dual chromatography column set up in tandem position with a convenient support rack. The top chromatography column was filled with Dowex-2 $\text{W} \times 8$, 200–400 mesh Cl^- form resin (74–37 μm) and the bottom column with Dowex-50 $\text{W} \times 8$, 200–400 mesh, H^+ form resin (74–37 μm). The top column was washed with 10 ml of water and allowed to drain through both the columns. Urine sample was applied to the top column. The top sample applied was passed through the resin followed by lower column resin and then eluted out. The top column was now washed three times with water and was allowed to drain through both columns like the urine sample. The top anionic columns were removed and bottom cationic columns are placed in empty, numbered test tubes, δ -ALA is eluted individually from each column with sodium acetate trihydrate (1.0 M) and collected in a test tube. 0.2 ml of acetyl acetone was added to test tubes and mixed. Test tubes were then placed in boiling water bath for 10 min and cooled to room temp. Added equal amount of freshly prepared Ehrlich reagent (10 g of *p*-dimethylaminobenzaldehyde in 300 ml of glacial acetic acid +80 ml 70% perchloric acid, and diluted to

500 ml with glacial acetic acid) mixed. Optical density was read at 555 nm after 15 min. The development of pink color indicates the presence of δ -ALA.

2.5.3. Clinical hematological variables

Levels of white blood cells (WBC), red blood cells (RBC), hemoglobin, hematocrit, MCV, MCH, and MCHC were measured on Sysmex hematology analyzer (model K4500).

2.5.4. Blood glutathione (GSH)

Analysis of blood GSH concentration was performed by modified method of Jollow et al. (1974). 0.2 ml whole blood was added to 1.8 ml distilled water and incubated for 10 min at 37°C for complete hemolysis. After adding 3 ml 4% sulpho salicylic acid, tubes were centrifuged at 2500 rpm for 15 min. To the supernatant, 0.2 ml of 10 mM solution of 5,5'-dithiobis- (2 nitro benzoic acid) (DTNB) was added in the presence of phosphate buffer (0.1 M pH 7.4). Absorbance recorded at 412 nm was used for calculation of GSH concentration.

2.5.5. Liver δ -aminolevulinic acid dehydratase (ALAD) (EC 4.2.1.24) and δ -aminolevulinic acid synthetase (ALAS) activity (2.3.1.37)

Liver δ -aminolevulinic acid dehydratase (ALAD) and δ -aminolevulinic acid synthetase (ALAS) activities were determined following the method of Goering et al. (1988). 20% tissue homogenate (w/v) was prepared in 0.05 M Tris-HCl (pH 7.4) and 0.25 M Sucrose. The incubation medium (0.3 ml) contained 4 mg protein, 50 mM glycine, 50 mM-sodium citrate, 25 mM-sodium monophosphate, 10 mM-magnesium chloride, 5 mM-pyridoxal 5-phosphate and 10 mM disodium EDTA. The mixture was incubated for 45 min at 37°C in darkness. The aminoketones formed were converted into pyrroles by condensation with acetyl acetone. The activity of the enzyme was determined by adding Ehrlich reagent and the absorbance was recorded at 555 nm after 5 min.

2.5.6. Tissue reduced (GSH) and oxidized (GSSG) glutathione level

Tissue GSH and GSSG estimation were performed as described by the method of Hissin and Hilf (1973). Briefly, 0.25 g of tissue sample was homogenized on ice with 3.75 ml 0.1 M phosphate – 0.005 M EDTA buffer (pH 8.0) and 1 ml of 25% HPO_3 which was used as a protein precipitant. The homogenate (4.7 ml) was centrifuged at 38,500 rpm for 30 min at 4°C . For the GSH assay, 0.5 ml supernatant and 4.5 ml phosphate buffer (pH 8.0) were mixed. The final assay mixture (2.0 ml) contained 100 μl supernatant, 1.8 ml phosphate – EDTA buffer and 100 μl *O*-phthaldehyde (OPT; 1000 $\mu\text{l}/\text{ml}$ in absolute methanol, prepared fresh). After mixing, fluorescence was determined at 420 nm with an excitation wavelength of 350 nm using a spectrofluorometer (Model RF 5000 Shimadzu, Japan).

For the GSSG assay, 0.5-ml supernatant was incubated at room temperature with 200 μl of 0.04 mol/L *N*-ethylmaleimide solution for 30 min. To this mixture, 4.3 ml of 0.1 mol/L NaOH was added. A 100- μl sample of this mixture was taken for the measurement of GSSG using the procedure described above for GSH assay, except that 0.1 mol/L NaOH was used as the diluents instead of phosphate buffer.

2.5.7. Glutathione peroxidase (GPx) (EC 1.11.1.9)

Glutathione peroxidase activity was measured by the procedure of Flohe and Gunzler (1984). Supernatant obtained after centrifuging 5% brain homogenate at $1500 \times g$ for 10 min followed by 11,000 rpm for 30 min at 4°C was used for GPx assay. One ml of reaction mixture was prepared which contained 0.3 ml of phosphate buffer (0.1 M, pH 7.4), 0.2 ml of GSH (2 mM), 0.1 ml of sodium azide (10 mM), 0.1 ml of H_2O_2 (1 mM) and 0.3 ml of brain supernatant. After incubation at 37°C for 15 min, reaction was terminated by addition of 0.5 ml 5% TCA. Tubes were centrifuged at $1500 \times g$ for 5 min and supernatant was collected. 0.2 ml phosphate buffer (0.1 M pH 7.4) and 0.7 ml of DTNB (0.4 mg/ml) was added to 0.1 ml of reaction supernatant. After mixing, absorbance was recorded at 420 nm.

2.5.8. Superoxide dismutase (SOD) (EC 1.15.1.1)

Tissue SOD activity was assayed by the method of Kakkar et al. (1984). Reaction mixture contained 1.2 ml (0.052 mM) sodium pyrophosphate buffer,

0.1 ml (186 μ M) phenazine methosulphate, 0.3 ml (300 μ M) nitro blue tetrazolium. Reaction was initiated by adding 0.2 ml NADH (780 μ M) and stopped by adding 1 ml glacial acetic acid. Color intensity of the chromogen was measured at 560 nm. Activity was expressed as units/min/mg of protein.

2.5.9. Catalase (EC 1.11.1.6)

Catalase activity in blood was assayed following the procedure of Sinha (1972). 0.1 ml of 5% RBC hemolysate was incubated with 0.5 ml of H_2O_2 (0.2 M) at 37 °C for 90 s precisely, in the presence of 0.01 M phosphate buffer (pH 7.4). Reaction was stopped by adding 5% dichromate solution. Further samples were incubated at 100 °C for 15 min in boiling water bath. Amount of H_2O_2 consumed was determined by recording absorbance at 570 nm.

2.5.10. Alanine aminotransferase (ALT) (EC 2.6.1.2) and aspartate aminotransaminases (AST) (EC 2.6.1.1)

Alanine aminotransferase (ALT) and aspartate aminotransferase (AST) activities were measured in liver as described by Reitman and Frankel (1957). The assay system contained 1 ml of buffer/substrate solution, 0.2 ml of liver homogenate (10%w/v) and was incubated for exactly 60 min (for ALT) and 30 min for AST at 37 °C in a water bath. 1 ml of chromogen solution was added, mixed and allowed to stand for 20 min at room temperature and 10 ml of 0.4N NaOH was added subsequently. The extinction was read at 505 nm against a blank. The controls were run in parallel, the substrate being added after deproteinization.

2.5.11. Alkaline phosphatase (ALP) (EC 3.1.3.1) and acid phosphatase (ACP) (EC 3.1.3.2)

The activities of alkaline (ALP) and acid phosphatase (ACP) liver tissues were determined as described by Moss (1984). A total of 0.5 ml of the homogenate was reacted with 0.1 ml of Triton X-100. A 4-ml quantity of alkaline or acid buffer was added to this and the mixture was incubated at 37 °C for 1 h. After incubation, 0.5 ml of 30% trichloroacetic acid (TCA) was added to stop the reaction. The mixture was centrifuged; 2 ml of the supernatant were mixed with 6.6 ml of distilled water, 1 ml of 2.5% ammonium molybdate and 0.4 ml of amino-naphthol sulfuric acid. Optical density was measured at 620 nm and the phosphorus liberated was calculated.

2.5.12. Thiobarbituric acid reactive substances (TBARS)

Tissue lipid peroxidation was measured by method of Onkawa et al. (1979). Tissue homogenate was incubated with 8.1% SDS (w/v) for 10 min followed by addition of 20% acetic acid (pH 3.5). Reaction mixture was incubated with 0.6% TBA (w/v) for 1 h in boiling water bath. Pink color chromogen so formed was extracted in butanol pyridine solution and read at 532 nm. The amount of TBARS was calculated using a molar extinction coefficient of 1.56×10^5 M per cm.

2.5.13. Liver metallothionein (MT)

Total MT concentration in liver and kidneys were measured following the method of Onosaka and Cherian (1982). Briefly, one gram liver or 0.5 gm kidney was homogenized in 4 volumes of 0.25 M sucrose and used for MT. For MT estimation the final volume was adjusted to 3.4 ml with 30 mM Tris–HCl buffer, pH 8.0, 1 ml cadmium chloride solution (0.088 mol/l, Cd) and incubated for 5 min at room temperature. Excess cadmium was removed and precipitated by the addition of rat RBC hemolysate and heat treatment in a water-bath for 1 min. The denatured Cd-bound hemoglobin was removed by centrifuging at $1000 \times g$ for 5 min at room temperature. The step was repeated three times. The color of the supernatant was slightly yellowish but transparent. The amount of Cd in the obtained supernatant fraction is a measure of MT-bound Cd and was determined using an atomic absorption spectrophotometer (AAS; Perkin Elmer model AAnalyst 100 USA). The concentration of MT in each tissue was calculated assuming that 6 atoms of Cd are bound to each molecule of thionein, which has a molecular weight of 6050 by amino acid analysis.

2.5.14. Reactive oxygen species (ROS)

Amount of ROS in tissue and blood was measured using 2', 7'-dichlorofluorescein diacetate (DCF-DA) that gets converted into highly fluorescent DCF by cellular peroxides (including hydrogen peroxide). The assay was

performed as described by Socci et al. (1999). Briefly, tissue was homogenized (10 mg) in 1 ml of ice-cold 40 mM Tris–HCl buffer (pH 7.4) and further diluted to 0.25% with the same buffer and placed on ice. For estimation of ROS in blood 5% RBC hemolysate was prepared and diluted to 1.5% with ice-cold 40 mM Tris–HCl buffer (pH 7.4). The samples were divided into two equal fractions. In one fraction 40- μ l of 1.25 mM DCF-DA in methanol was added for ROS estimation. Another fraction in which 40- μ l of methanol was added, served as a control for tissue/hemolysate auto fluorescence. All samples were incubated for 15 min in a 37 °C water bath. Fluorescence was determined at 488 nm excitation and 525 nm emission using a fluorescence plate reader (Tecan Spectra Fluor Plus).

2.5.15. Metal estimation

Arsenic concentration in blood, liver and kidneys were measured after wet acid digestion using a Microwave Digestion System (CEM, USA, model MDS-2100). Arsenic was estimated using a Hydride Vapour Generation System (Perkin Elmer model MHS-10) fitted with an atomic absorption spectrophotometer (AAS, Perkin Elmer model AAnalyst 100). Zinc, copper and magnesium contents were also measured in the digested tissue samples using AAS (Parker et al., 1967).

2.5.16. IR spectra

Infrared spectra were recorded as KBr pellet using a Perkin-Elmer FT-IR spectrophotometer and infrared spectra have been determined in the 4000–400 cm^{-1} region.

2.5.17. Statistical analysis

Data are expressed as mean $\pm$ SEM. Data comparisons were carried out using one-way analysis of variance (ANOVA) followed by Bonferroni's test to compare means between the different treatment groups. Difference between unexposed and exposed (with or without treatment) with a p value <0.05 was considered significant.

3. Results

3.1. Effects on blood and tissue reactive oxygen species (ROS) level

Blood and tissue ROS levels increased significantly in animals exposed to arsenic, while administration of *M. oleifera* in normal animals had no effect (Fig. 1). Concomitant administration of *M. oleifera* seed powder with arsenic provided significant protection to reactive oxygen species (ROS) levels in blood (57% by 500 mg/kg dose of *M. oleifera*) and tissues (56% in liver, 42% in kidneys and brain). The protection was more pronounced in kidney at the dose of 500 mg/kg compared to 250 mg, while in blood, liver and brain there was no difference between the two doses.

3.2. Effects on heme synthesis pathway in blood and liver

Blood and liver ALAD activity showed significant inhibition (69% in blood and 73% in liver) on arsenic exposure accompanied by an elevated urinary excretion of ALA (Fig. 2). Administration of *M. oleifera* seed powder in normal animals had no effect on these variables. However, concomitant administration of 500 mg/kg of *M. oleifera* seed powder with arsenic provided significant protection in blood (25% depletion in the activity compared to normal animals) and liver ALAD activity (17% inhibition in the activity compared to normal animal) while, no protection was offered by the lower dose. Both the

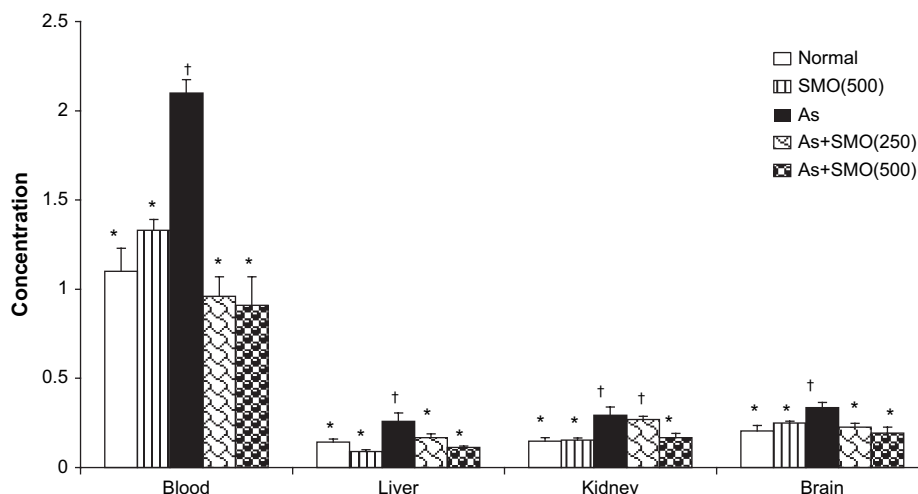


Fig. 1. Co-administration of *Moringa oleifera* with arsenic on the level of reactive oxygen species (ROS) in blood, liver, kidney and brain of male mice. *Abbreviation used and units*: ROS – η moles $\text{min}^{-1} \text{ml}^{-1}$ blood; ROS – η moles $\text{min}^{-1} \text{mg}^{-1}$ protein in tissues. Values are mean $\pm$ SE; $n = 5$. *[†]Differences between values with matching symbol notations within each block of histogram are not statistically significant at 5% level of probability.

doses of *M. oleifera* however, offered significant decrease in urinary ALA excretion in arsenic co-exposed mouse compared to arsenic exposed control group (Fig. 2).

3.3. Effect on some hematological variables of blood

The effects of oral administration of *M. oleifera* seed powder on GSH level, catalase activity and few other hematological variables are shown in Table 1. Blood GSH level and catalase activity remained statistically unchanged on arsenic exposure except for a moderate depletion in their level. On the other hand, platelet counts showed a significant elevation on arsenic exposure. Animals concomitantly exposed to *M. oleifera* and arsenic showed normal GSH level (2.84 ± 0.05 in arsenic plus 500 mg/kg *M. oleifera* treated animals compared to 3.15 ± 0.05 in normal animals) and catalase activity (4.95 ± 0.18 in arsenic plus 500 mg/kg *M. oleifera* treated animals compared to 5.12 ± 0.03 in normal animals) while,

significantly decreased platelet counts (particularly at the dose of 500 mg/kg). The other hematological variables remained insensitive to any of the treatments.

3.4. Effect on hepatic enzymes of antioxidant defense system

Exposure to arsenic led to a significant depletion in the activities of SOD (36%), catalase (8%) and GPx (20%) in mouse liver (Table 2). These changes were accompanied by a significant increase (67%) in thiobarbituric acid reactive substances (TBARS). No change however was observed in the activity of liver ALAS on arsenic exposure. Administration of *M. oleifera* seed powder too had no effect on these variables in normal animals. On the other hand, co-administration of *M. oleifera* (particularly at the higher dose), with arsenic provided significantly protection to the above mentioned biochemical alterations.

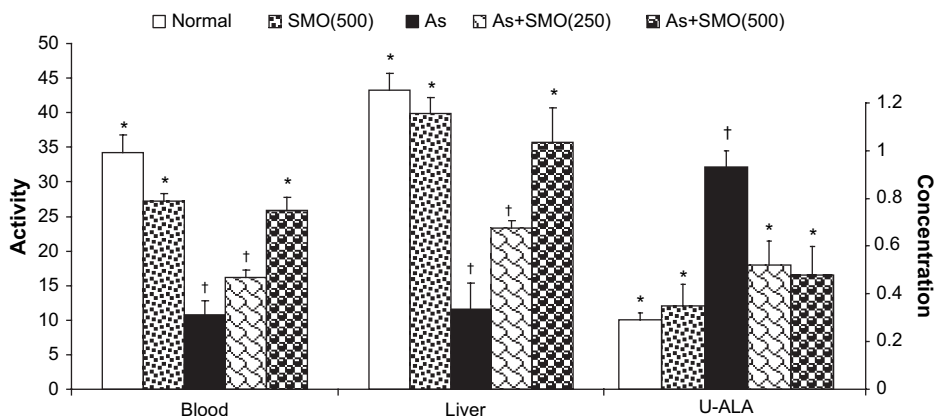


Fig. 2. Co-administration of *Moringa oleifera* with arsenic on blood and liver variables indicative of altered heme biosynthesis of male mice. *Abbreviation used and units*: ALAD – δ -aminolevulinic acid dehydratase as nmol/min/ml erythrocytes in blood; ALAD – δ -aminolevulinic acid dehydratase as nmol/min/mg protein in liver. Values are mean $\pm$ SE; $n = 5$. *[†]Differences between values with matching symbol notations within each block of histogram are not statistically significant at 5% level of probability.

Table 1
Co-administration of *Moringa oleifera* with arsenic on some biochemical variables in blood of male mice

	Normal animals	SMO (500 mg)	As	As + SMO (250 mg)	As + SMO (500 mg)
GSH	3.15 ± 0.05*	3.00 ± 0.09*	2.54 ± 0.03 [†]	2.72 ± 0.17 [†]	2.84 ± 0.06 [†]
Catalase	5.12 ± 0.03*	4.87 ± 0.54*	4.29 ± 0.09 [†]	4.45 ± 0.32 [†]	4.95 ± 0.18*
RBC	8.35 ± 0.17*	7.69 ± 0.26*	7.95 ± 0.17*	8.40 ± 0.34*	8.21 ± 0.41*
WBC	8.35 ± 0.45*	8.10 ± 0.20*	10.25 ± 2.65*	9.35 ± 0.71*	8.82 ± 1.95*
Hb	12.7 ± 0.32*	12.2 ± 0.50*	12.5 ± 0.40*	12.8 ± 0.35*	12.7 ± 0.45*
HCT	41.2 ± 0.95*	38.9 ± 1.00*	39.3 ± 0.70*	41.9 ± 1.29*	41.0 ± 1.60*
PLT	592 ± 5.00*	604 ± 51.50*	838 ± 25.00 [†]	929.8 ± 58.6 [†]	426.6 ± 39.5*

Abbreviation used and units: GSH – glutathione as mg/ml; Catalase as μ moles of H_2O_2 consumed/min/ml; RBC – red blood cells as $\times 10^6/\mu$ l; WBC – white blood cells as $\times 10^3/\mu$ l; Hb – hemoglobin as gm/dL; HCT – hematocrit as %; PLT – platelet count as $10^3/\mu$ l. Values are mean $\pm$ SE; $n = 5$. *[†]Differences between values with matching symbol notations within each column are not statistically significant at 5% level of probability.

3.5. Effects on liver injury

A significant depletion of hepatic ACP, ALP and AST activities in arsenic exposed mice suggest liver injury (Table 3). Concomitant administration of *M. oleifera* seed powder with arsenic provided significantly protection to these altered liver biochemical variables.

3.6. Effects on kidney oxidative stress

A significant depletion in catalase activity (47%) and a significant increase (23%) in TBARS level were observed on arsenic exposure (Table 4). Kidney SOD, GSH and GSSG levels however, remained unchanged. Co-administration of *M. oleifera* (particularly at the dose of 500 mg/kg) with arsenic provided significant recovery in the altered catalase activity (62%) and TBARS levels (17%) towards normal value.

3.7. Effects on brain oxidative stress

Exposure to arsenic led to a significant depletion in brain SOD activity (32%) while TBARS level (27%) showed significant increase on arsenic exposure in mouse (Table 5). Brain GSH and GSSG levels remained insensitive to arsenic or *M. oleifera* either individually or in combination. Co-administration of *M. oleifera* significantly protected SOD activity (19%) and reduced increase in TBARS level (18%) in arsenic co-exposed animals.

3.8. Effect on liver metallothionein

Fig. 3 shows a significant induction of hepatic metallothionein (MT) (182%) in arsenic exposed mouse liver. Administration of *M. oleifera* seed powder in normal animals did not elicit any change in MT level. Co-administration of *M. oleifera* with arsenic however, significantly reduced (73% by 500 mg/kg dose and 64% by the lower dose) hepatic MT levels and the values were near to the normal animals.

3.9. Effect on arsenic and essential metal concentration

Fig. 4 shows arsenic concentration in blood, liver, kidneys and brain of arsenic exposed mice treated with *M. oleifera* seed powder. Arsenic exposure significantly increased its concentration in these tissues. Animals co-administered with arsenic and *M. oleifera* showed marked depletion of arsenic concentration in blood (55%), liver (65%) and kidney (54%). The changes were more pronounced at 500 mg/kg dose of *M. oleifera* than at 250 mg/kg. No significant change in brain arsenic concentration was noted after the administration of *M. oleifera* seed powder.

No prominent changes in blood and tissue copper, zinc and iron concentration in arsenic exposed animals either alone or in combination with *M. oleifera* seed powder was noted (Figs. 5–7).

4. Discussion

The present study evaluates the effects of crude powder of *M. oleifera* seeds on arsenic-induced oxidative stress, hepatic

Table 2
Co-administration of *Moringa oleifera* with arsenic on some hepatic biochemical variables suggestive of oxidative stress in male mice

	Normal animals	SMO (500 mg)	As	As + SMO (250 mg)	As + SMO (500 mg)
SOD	1.51 ± 0.15*	1.45 ± 0.17*	0.97 ± 0.03 [†]	1.36 ± 0.15*	1.77 ± 0.16*
Catalase	7.45 ± 0.16*	8.03 ± 0.72*	6.89 ± 0.12 [†]	6.83 ± 0.85 [†]	7.78 ± 0.21*
GPx	5.07 ± 0.28*	4.56 ± 0.43*	4.05 ± 0.05 [†]	5.16 ± 0.34 [†]	5.67 ± 0.27*
ALAS	6.19 ± 0.24*	7.28 ± 0.96*	5.07 ± 0.53*	5.40 ± 0.51*	5.07 ± 0.15*
TBARS	4.92 ± 0.04*	5.52 ± 0.20*	8.20 ± 0.84 [†]	5.48 ± 0.20*	4.40 ± 0.19*

Abbreviation used and units: SOD – super oxide dismutase as units/min/mg protein; Catalase as μ moles of H_2O_2 consumed/min/mg protein; GPx – μ g/min/mg protein; ALAS – δ -aminolevulinic acid synthetase as nmol/min/mg protein; TBARS – thiobarbituric acid reactive substances as μ g/g of tissue. Values are mean $\pm$ SE; $n = 5$. *[†]Differences between values with matching symbol notations within each column are not statistically significant at 5% level of probability.

Table 3

Co-administration of *Moringa oleifera* with arsenic on some hepatic biochemical variables suggestive of liver damage in male mice

	Normal animals	SMO (500 mg)	As	As + SMO (250 mg)	As + SMO (500 mg)
ACP	0.107 ± 0.009*	0.097 ± 0.005*	0.075 ± 0.003 [†]	0.072 ± 0.007*	0.088 ± 0.005*
ALP	0.043 ± 0.008*	0.034 ± 0.002*	0.021 ± 0.003 [†]	0.035 ± 0.002 [§]	0.045 ± 0.004*
ALT	6.871 ± 0.790*	6.671 ± 0.100*	6.736 ± 1.100*	5.492 ± 0.680*	5.780 ± 0.780*
AST	14.329 ± 1.160*	12.935 ± 0.371*	11.082 ± 0.700 [†]	9.313 ± 0.390 [†]	8.921 ± 0.320 [†]

Abbreviations and units: ACP — acid phosphatase as mgP/hr/mg of protein; ALP — alkaline phosphatase as mgP/gm/mg of protein; AST and ALT — aspartate and alanine aminotransferase as nmol/min/mg of protein. Values are mean ± SE; $n = 5$. *,[†],[§]Differences between values with matching symbol notations within each column are not statistically significant at 5% level of probability.

MT and on the concentration of arsenic and few essential metals in blood and other soft tissues. It is evident from the present study that co-administration of *M. oleifera* seed powder with arsenic provided significant protection to arsenic-induced oxidative stress and also prevented accumulation of arsenic in blood and other soft tissues.

The arsenic toxicity is mediated via generation of reactive oxygen species (ROS) (Nishigori et al., 2004), which in turn lead to the development of carcinogenesis and other cytotoxic effects (Wang and Huang, 1994). Superoxide anion, hydroxyl radical and hydrogen peroxides are the intermediates formed during oxygen metabolism (Saxena and Flora, 2004). Exposure to arsenic produced a significant depletion in the activity of blood ALAD. ALAD is a sulfhydryl containing enzyme that catalyzes the asymmetric condensation of two molecules of aminolevulinic acid (ALA — substrate) to porphobilinogen during heme synthesis pathway. Since arsenic has a high affinity for sulfhydryl, this may lead to possible inhibition of GSH concentration and ALAD activity. An inhibition in the activity of blood ALAD may lead to a significant accumulation of ALA. Therefore there is a possibility that an increased ALA might be generating reactive oxygen intermediates (ROIs) and oxidative stress (Hermes-Lima, 1995; Bhadauria and Flora, 2003). Significant recovery in blood and liver ALAD activity, GSH and platelets counts, urinary ALA excretion besides a significant depletion of ROS production following co-administration of *M. oleifera* in arsenic exposed animals are interesting observations. These beneficial effects may be due to the direct scavenging effect of *M. oleifera*. *M. oleifera* seeds contain significantly high concentration of methionine and cysteine. Due to the presence of -SH group, these amino acids stimulate GSH synthesis and provide additional binding site for arsenic that ultimately facilitate the removal of arsenic

from the body. Oxidative methylation reactions in which trivalent forms of arsenic is sequentially methylated to form mono- and di-methylated products (MMA, DMA, excretory products) using S-adenosyl methionine (SAM) as the methyl donor and GSH as an essential co-factor (Vahter, 2002). S-adenosyl methionine, the universal methyl donor is synthesized from the condensation of L-methionine with ATP. L-methionine thus might indirectly be playing an important role in the removal of arsenic from the *in vivo* site. The possible interaction between cysteine and arsenic too could be represented as shown in the Fig. 8.

Reduced glutathione (GSH), a non-protein thiol, plays a crucial role in protecting cells against reactive oxygen species (Ito et al., 1998). It also plays an important role in the detoxification via facilitating removal of arsenic from the cellular sites and stimulating excretion of methylated arsenic (Vahter, 2002), which is responsible for arsenic induced cytotoxicity. GSH depletion may result in the accumulation of free radicals that may initiate lipid peroxidation (LPO), resulting in biochemical injury by covalent binding to bio-macromolecules (Oliveira et al., 1999).

Two potential cellular defense mechanisms against oxidative stress are the low molecular weight, cysteinyl-containing polypeptide and tripeptide known as metallothionein and GSH. Metallothionein (MT) are low molecular weight (<10 KDa) metal binding proteins involved in the resistance to heavy metals and metal homeostasis (Hamer, 1986). It has been reported earlier that arsenic (III) might increase the level of hepatic MT in rats (Flora and Tripathi, 1998; Albores et al., 1992). Due to its high cysteine contents, MT binds avidly with number of metals leading to the formation of metal-MT complex. Interestingly, co-administration of *M. oleifera* with arsenic led to a significant depletion of hepatic MT, which could

Table 4

Co-administration of *Moringa oleifera* with arsenic on some kidney biochemical variables suggestive of oxidative stress in male mice

	Normal animals	SMO (500 mg)	As	As + SMO (250 mg)	As + SMO (500 mg)
Catalase	10.91 ± 0.20*	8.76 ± 0.035*	5.73 ± 1.13 [†]	6.45 ± 0.35 [†]	9.28 ± 0.83*
SOD	2.44 ± 0.22*	3.62 ± 0.14*	3.31 ± 0.14*	2.69 ± 0.54*	2.86 ± 0.062*
TBARS	5.92 ± 0.36*	5.48 ± 0.36*	7.28 ± 0.14 [†]	6.72 ± 0.18 [†]	6.02 ± 0.12*
GSH	1.14 ± 0.020*	0.90 ± 0.01*	1.12 ± 0.031*	0.92 ± 0.011*	0.95 ± 0.014*
GSSG	0.128 ± 0.002*	0.134 ± 0.002*	0.125 ± 0.002*	0.131 ± 0.001*	0.129 ± 0.001*
GSH:GSSG	8.9	7.2	8.35	7.36	7.89

Abbreviation used and units: Catalase as μ moles of H_2O_2 consumed/min/mg protein; SOD — super oxide dismutase as units/min/mg protein; TBARS — thiobarbituric acid reactive substances as μ g/g of tissue; GSH — reduced glutathione as mg/gm tissue; GSSG — oxidized glutathione as mg/gm tissue. Values are mean ± SE; $n = 5$. *,[†],[§]Differences between values with matching symbol notations within each column are not statistically significant at 5% level of probability.

Table 5

Co-administration of *Moringa oleifera* with arsenic on some biochemical variables suggestive of brain oxidative stress in male mice

	Normal animals	SMO (500 mg)	As	As + SMO (250 mg)	As + SMO (500 mg)
SOD	7.83 ± 0.24*	7.01 ± 0.85*	5.30 ± 0.54 [†]	7.65 ± 1.19 [§]	6.33 ± 0.14 [†]
TBARS	5.38 ± 0.08*	5.61 ± 0.10*	6.84 ± 0.18 [†]	7.44 ± 0.18 [†]	5.64 ± 0.34*
GSH	1.11 ± 0.012*	1.15 ± 0.005*	1.16 ± 0.030*	1.11 ± 0.057*	0.92 ± 0.010*
GSSG	0.115 ± 0.002*	0.114 ± 0.001*	0.116 ± 0.0005*	0.112 ± 0.001*	0.117 ± 0.003*
GSH:GSSG	9.65	10.17	9.91	9.91	7.86

Abbreviation used and units: SOD — super oxide dismutase as units/min/mg protein; TBARS — thiobarbituric acid reactive substances as µg/g of tissue; GSH — reduced glutathione as mg/gm tissue; GSSG — oxidized glutathione as mg/gm tissue. Values are mean ± SE; n = 5. *[†][§]Differences between values with matching symbol notations within each column are not statistically significant at 5% level of probability.

be correlated with arsenic elimination from the liver. Lipid is considered as an index to monitor the function of membrane integrity, while MDA (the end product of LPO) is measured as the index of lipid peroxidation (Ikuo et al., 1991). In the present investigation, there was a marked increase in TBARS levels in the liver, kidney and brain of arsenic exposed mice which were substantially reduced co-administration of *M. oleifera* seed powder. The preventive properties of *M. oleifera* seed powder have been related to the inhibition of lipid peroxides formation or scavenging of free radicals as evident from the decreased TBARS level. *M. oleifera* possesses numerous antioxidants including vitamin C, β-carotene (Nambiar and Seshadri, 2001), α- and γ-tocopherol (Sánchez-Machado et al., 2005), β-sitosterol, vitamin A (Nambiar et al., 2003) and phenolic compounds such as quercetin and kaempferol, flavonoids, anthocyanins (Siddhuraju and Becker, 2003). These components seem to offer protection and to maintain the membrane integrity of soft tissues. Rich source of ascorbic acid in pods and leaves of *M. oleifera* is known for its antioxidant properties and its beneficial effects could be attributed to its ability to form a poorly ionized but soluble complex with toxic metal/metalloid (Siddhuraju and Becker, 2003; Flora, 2002).

Antioxidants enzymes are considered to be body's primary defense, which prevents biological macromolecules from oxidative injury and removes peroxides, free radicals and superoxide anion generated with in the cell. Glutathione peroxidase (GPx) and catalase are the major enzymes that remove hydrogen peroxide generated by superoxide dismutase in cytosol and mitochondria by oxidizing GSH to GSSG (Park et al., 1994). We observed a significant depletion of GPx, catalase and superoxide dismutase (SOD) activities in tissues following arsenic exposure. The decrease in SOD activity may be attributed to enhanced superoxide radical production during arsenic metabolism (Yamanaka et al., 1989). The increase in superoxide radicals inhibits catalase activity (Kono and Fridovich, 1982). The paucity of NADPH production during arsenic exposure also decreases catalase activity (Kono and Fridovich, 1982; Kirkman and Gactani, 1984). Glutathione peroxidase (GPx) is a Se-dependent enzyme and it is known that arsenic binds with Se and form As-Se complex (Flora et al., 1999) resulting in the inhibition of GPx activity. In the present study, administration of *M. oleifera* significantly protected SOD, catalase and GPx activities by directly scavenging ROS as well as by inhibiting lipid peroxidation, suggesting antioxidant properties of *M. oleifera* seed powder.

One of the most interesting observations in the present study has been the capability of *M. oleifera* to reduce arsenic uptake in the target tissues. These results thus point to moderate chelating property of *M. oleifera* against arsenic, which is in agreement with few earlier reported studies describing antioxidant and chelating properties of *M. oleifera* *in vitro* and *in vivo* (Santos et al., 2005; Reddy et al., 2005; Gupta et al., 2005; Kumar and Pari, 2003).

As evident from the present results, *M. oleifera* seed powder showed some beneficial effects in depleting arsenic from the *in vivo* sites. It is although not yet clear which constituent(s) might be playing a role in the removal of arsenic. Arsenic exists in anionic and neutral forms while, oxyanions of arsenic exist in three different arsenite species viz. H_3AsO_3 , H_2AsO_3^- and HASO_3^{2-} in the pH range of 7–8, 10–11 and 12–13 respectively (Wasay et al., 1996). In the present study aqueous solution of *M. oleifera* was used, which is a heterogeneous complex mixture having cationic polypeptides with various functional groups particularly of low molecular weight amino acids (Jose et al., 1999). Thus, there is a possibility that the proteineous amino acids with variety of structurally related pH-dependent properties of generating either positively or negatively charge sites may attract the anionic and cationic species of arsenic (Costa et al., 1997). With the increase in pH range, the carboxylic group of the amino acids would

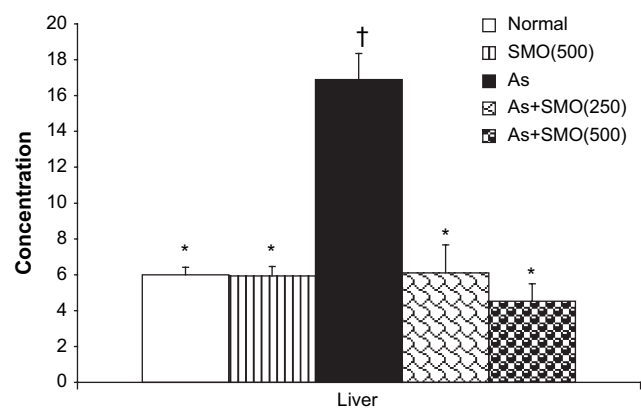


Fig. 3. Co-administration of *Moringa oleifera* with arsenic on liver metallothionein level of male mice. Abbreviation used and units: MT — Metallothionein as µg MT/gm of tissue. Values are mean ± SE; n = 5. *[†]Differences between values with matching symbol notations within each block of histogram are not statistically significant at 5% level of probability.

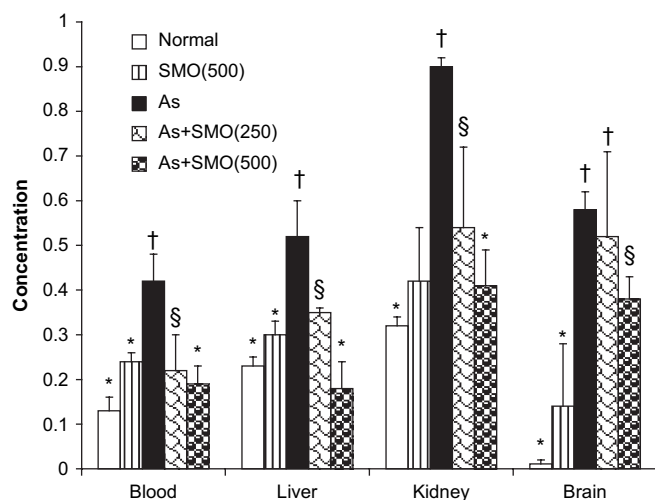


Fig. 4. Arsenic concentration in blood, liver, kidney and brain of arsenic exposed mice concomitantly administered the seed powder of *Moringa oleifera*. Units: Blood arsenic as ng/dl; liver, kidney and brain arsenic as ng/gm wet tissue. Values are mean $\pm$ SE; $n = 5$. *†§ Differences between values with matching symbol notations within each block of histogram are not statistically significant at 5% level of probability.

progressively be deprotonated as carboxylate ligands thus simultaneously protonating the amino group. Possibility of protein/amino acid-arsenic interaction can further be supported by IR spectrum of *M. oleifera* seed powder alone and in combination with arsenic (Fig. 9). We observed a shifting of NH_3^+ symmetric stretching bands at 2628.52 cm^{-1} and 2922.50 cm^{-1} and NH_3^+ asymmetric stretching band at 1652.14 cm^{-1} . The absence of characteristic peaks of carboxylate ions at $1600\text{--}1590\text{ cm}^{-1}$ further confirms the cationic nature of polypeptides present in *M. oleifera* seed powder. Such positively charged NH_3^+ ions facilitate the *M. oleifera*-arsenic complexation and

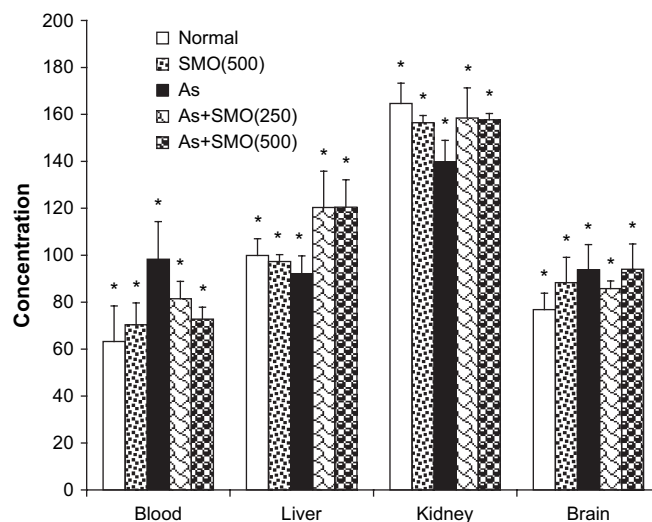


Fig. 6. Iron concentration in blood, liver, kidney and brain of arsenic exposed mice concomitantly administered the seed powder of *Moringa oleifera*. Units: Blood iron as $\mu\text{g/ml}$; liver, kidney and brain iron as $\mu\text{g/gm}$ wet tissue. Values are mean $\pm$ SE; $n = 5$. * Differences between values with matching symbol notations within each block of histogram are not statistically significant at 5% level of probability.

might be responsible for excretion of arsenic from *in vivo* sites. In order to suggest which co-ordinating/interacting site of a given ligand is responsible for binding with arsenic, methionine was reacted with sodium arsenite *in vitro* at equimolar ratio at room temperature for 60 min. IR spectra of methionine, sodium arsenite and reaction mixture (after evaporation of dryness in vacuum in rota vapour at $50\text{--}60^\circ\text{C}$) were recorded. No significant shift in amine (2948 and 2914 cm^{-1}) and carboxyl function (1649 and 1581 cm^{-1}) of methionine was observed before and after reaction (IR spectra not shown). This indicates

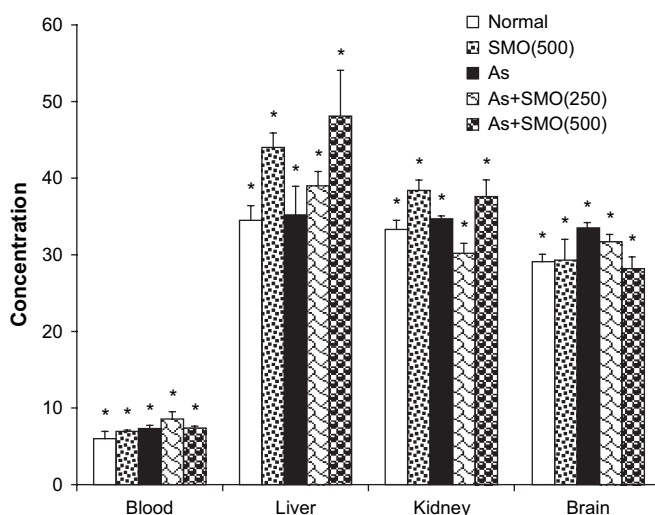


Fig. 5. Zinc concentration in blood, liver, kidney and brain of arsenic exposed mice concomitantly administered the seed powder of *Moringa oleifera*. Units: Blood zinc as $\mu\text{g/ml}$; liver, kidney and brain zinc as $\mu\text{g/gm}$ wet tissue. Values are mean $\pm$ SE; $n = 5$. * Differences between values with matching symbol notations within each block of histogram are not statistically significant at 5% level of probability.

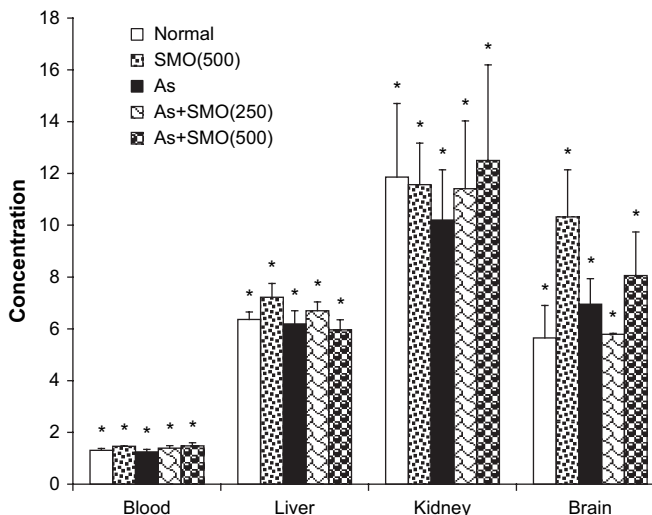


Fig. 7. Copper concentration in blood, liver, kidney and brain of arsenic exposed mice concomitantly administered the seed powder of *Moringa oleifera*. Units: Blood copper as $\mu\text{g/ml}$; liver, kidney and brain copper as $\mu\text{g/gm}$ wet tissue. Values are mean $\pm$ SE; $n = 5$. * Differences between values with matching symbol notations within each block of histogram are not statistically significant at 5% level of probability.

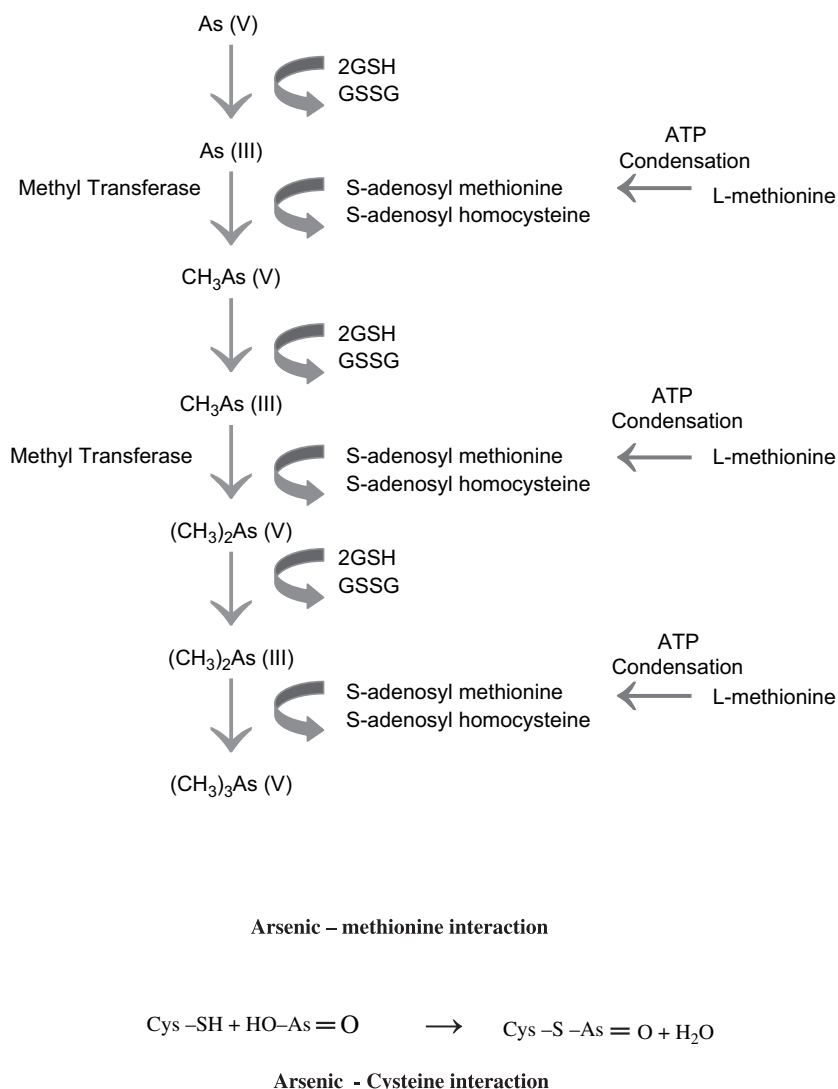


Fig. 8. Possible scheme showing interaction between arsenic and cysteine/methionine.

that amine and carboxyl function were not responsible for interaction of methionine with arsenic. However, it is interesting to note that arsenic oxygen stretching frequencies of sodium arsenite were significantly shifted from approximately 700 to 600 cm^{-1} . Probably, the sulfur of methionine interacts with arsenic causing this shift by changing the co-ordination number of arsenite.

Reaction of *M. oleifera* seed powder with arsenite in water was carried out at weight ratio of 1:0.006 (based on the dose of *M. oleifera* extract versus arsenite for *in vivo* experiments). The mixture was dried in rota vapour and IR spectra of *M. oleifera* seed powder and reaction mixture were recorded (Fig. 9). No significant change in spectra of reaction mixture and *M. oleifera* extract was observed. The amount of arsenite taken for the reaction was far less to give any significant change in IR spectra of *M. oleifera* extract. Nevertheless, based on the reaction of methionine with arsenite, it could be suggested that the arsenite interact with the ligands via S-As co-ordination.

In addition to the above, there are reports that suggest that the detoxified seeds of *M. oleifera* are rich in magnesium,

sodium, phosphorus and calcium (Anhwange et al., 2004), protein, and high calcium might act as an antagonistic to metals/metalloids (Flora, 2002). Santos et al. (2005) recently reported that *M. oleifera* seed powder possesses 20,000 MW glycoprotein, main native acidic protein, which has hemagglutinating activity. It might be possible that this glycoprotein binds with arsenic *in vivo* and thus forms a complex that could be excreted from the body. We thus recommend more detailed studies particularly related to the mechanism of action of *M. oleifera* in preventing arsenic toxicity.

5. Conclusions

It can thus be concluded from the present study that concomitant administration of *M. oleifera* seed powder with arsenic may have some beneficial role in protecting animals from the toxic effects of arsenic based on the biochemical parameters selected by us which are suggestive of tissue oxidative stress, alterations in heme synthesis pathway, organ damage and also in reducing arsenic uptake in target tissues.

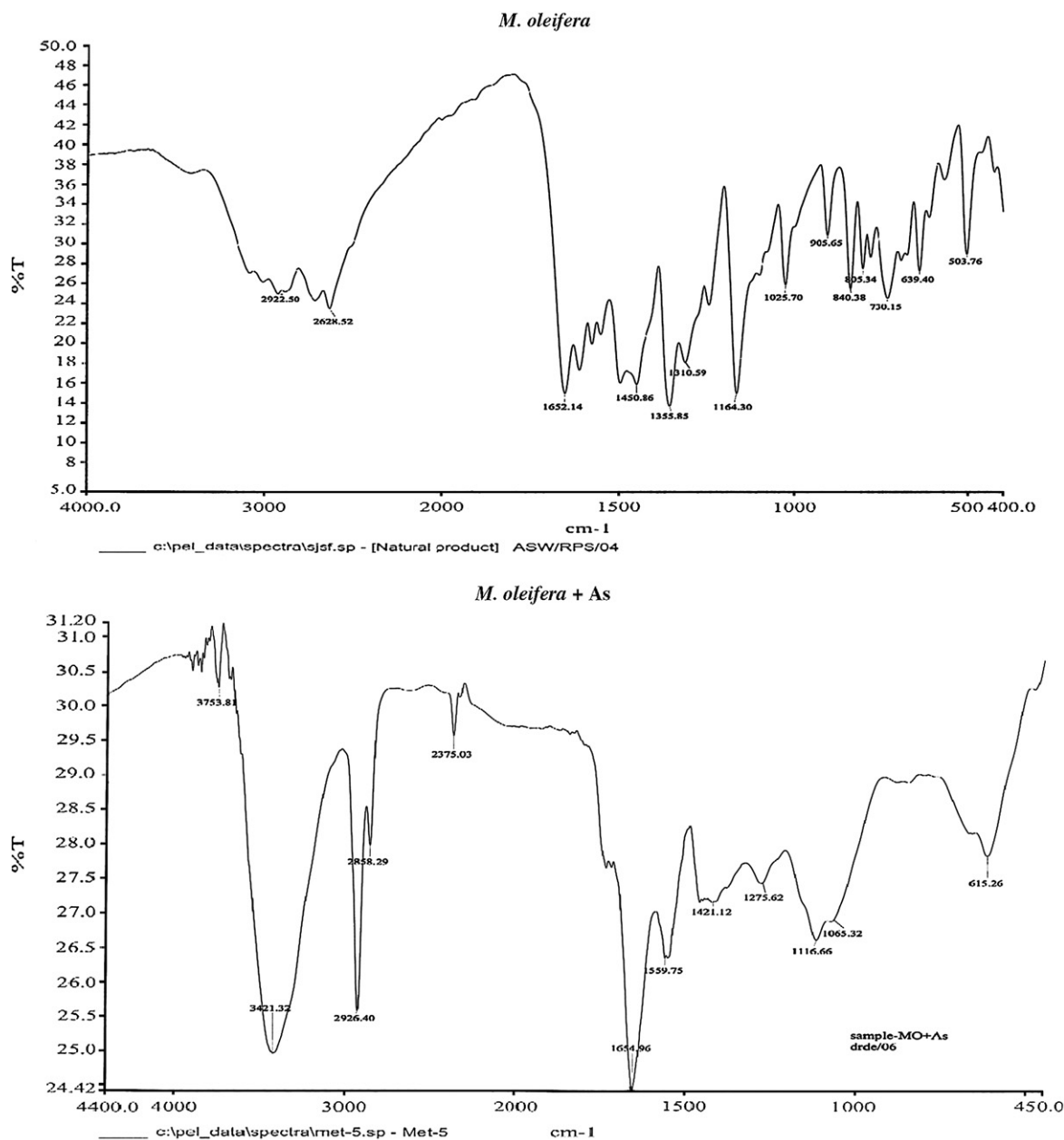


Fig. 9. Infrared Spectra of *M. oleifera* seed powder and arsenic + *M. oleifera*.

The mechanism of such protection however is still not clear but it could be due to the interaction between cysteine and methionine rich proteins that are present in high amounts in *M. oleifera* seeds and might be responsible for decreasing arsenic from *in vivo* sites. Beside this, *M. oleifera*, which is also rich with other potent antioxidants like vitamin C, vitamin E, β -carotene etc., might also be involved in combating arsenic toxicity in animals.

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